SE Lab 3

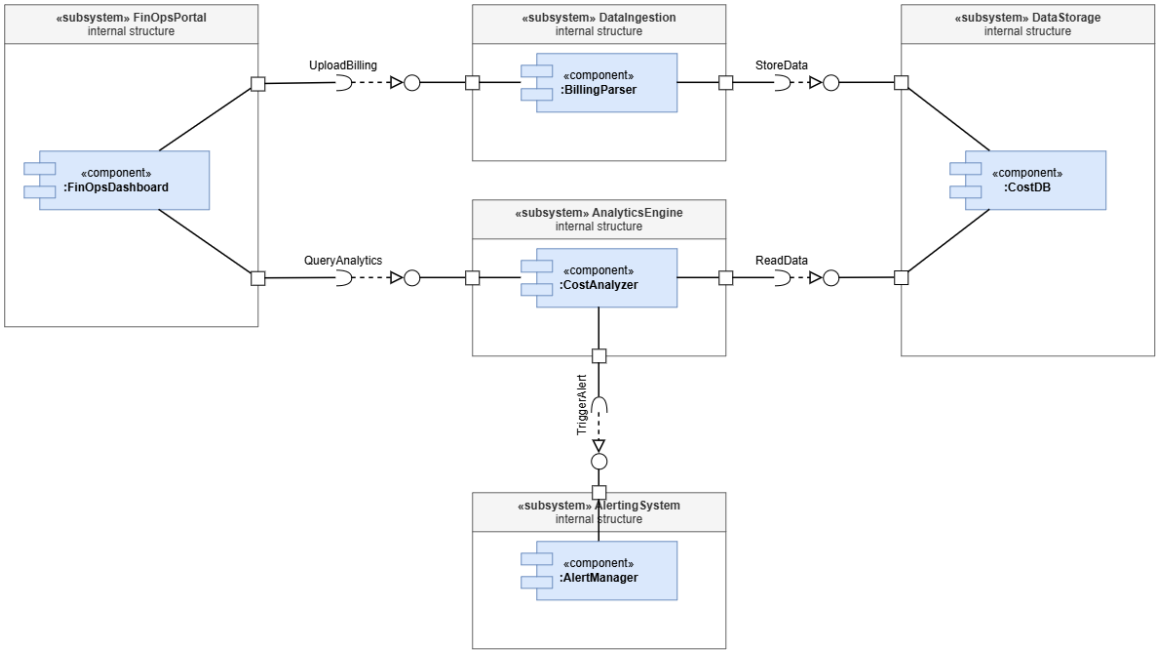
Component Modelling & Architectural Pattern Selection

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Component Diagram:



Architecture Selection

We chose Microservices Architecture for the Multi-Cloud Asset Cost Optimization Portal System.

Architectural Choice

Microservices Architecture

Reasons:

- Independent Scalability for Ingestion: The system is required to parse multi-gigabyte cloud billing files (AWS, Azure, GCP). By using microservices, the DataIngestion component can be scaled independently during peak periods (e.g., end-of-month billing cycles) to handle massive file sizes without dedicating unnecessary resources to the dashboard or alerting components.
- Separation of Concerns & Technology Diversity: The portal features distinct functional requirements, such as a FinOps Dashboard (requiring a lightweight web framework) and a CostAnalyzer engine (requiring heavy data processing to calculate CPU utilization over 14 days and flag idle instances). Microservices allow each component to utilize the most optimal technology stack for its specific task.

Security Advantage

- Fault Isolation and Scoped Access: Microservices inherently limit the attack surface by isolating components. The DataStorage and DataIngestion services can be placed in a private subnet with strict Identity and Access Management (IAM) policies. If the FinOps Portal (front-end) is compromised, the attacker does not automatically gain direct access to the centralized CostDB or the sensitive billing CSVs, as data access is strictly gated through authenticated API interfaces.

Performance Benefit

- Asynchronous Processing and Reduced Latency: To meet the non-functional requirement (NFR-001) of parsing multi-gigabyte files and updating cost dashboards in under 2 minutes, a microservices architecture enables asynchronous, parallel processing. The BillingParser can stream data directly to the CostAnalyzer using message queues, preventing the web dashboard from blocking or timing out while heavy computations (e.g., identifying < 5% CPU utilization) are executed in the background.